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An Autonomous Institution

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
(ARTIFICIAL INTELLIGENCE & MACHINE LEARNING)

INNOVATIVE ASSIGNMENT – I

FOUNDATIONS OF ARTIFICIAL INTELLIGENCE

PROJECT REPORT

CAMPUS ROUTE FINDER USING BREADTH-FIRST SEARCH (BFS)

Course Code/Title: 25ML35T – Foundations Of Artificial Intelligence

Batch No.: B6

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Date: 28.07.2026

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1. ABSTRACT

The Campus Route Finder is a graph-based application that finds the shortest route between two locations on a campus using Breadth-First Search (BFS). Each campus location is represented as a vertex and each direct connection as an edge. Since the graph is unweighted, BFS explores locations level by level and finds a path with the minimum number of edges. The system displays the shortest route and its hop count.

2. INTRODUCTION

A graph is a data structure consisting of vertices and edges that can be used to represent connections between locations. In a campus environment, buildings such as classrooms, the library, canteen and laboratory can be represented as vertices, while direct routes between them can be represented as edges.

Breadth-First Search (BFS) is a graph traversal technique that explores all neighboring vertices before moving to the next level. For an unweighted graph, BFS can find a shortest path measured by the fewest number of edges. This makes it suitable for a campus route-finding problem where each direct connection is treated equally.

3. PROBLEM STATEMENT

Students and visitors may find it difficult to determine the shortest route between two locations on a large campus.

Proposed Solution:

Develop a Campus Route Finder using BFS that takes a starting location and destination, searches the campus graph, and displays the shortest route along with the number of hops.

4. OBJECTIVES

- Represent campus locations using a graph.
- Implement Breadth-First Search (BFS).
- Find the shortest route between two locations.
- Display the path clearly.
- Calculate the hop count.
- Demonstrate why BFS gives the fewest-edge path.

5. AGENT COMPONENTS

Component	Description
Environment	Campus locations and connecting routes
Percepts	Starting location, destination, and available connections
Agent	Campus Route Finder using BFS
Actions	Visit neighboring campus locations
Goal	Find the shortest route between two locations
Performance Measure	Minimum number of hops (edges)

6. METHODOLOGY

The proposed system receives campus locations, their connections, a starting location and a destination. The campus is modeled as an unweighted graph and BFS is applied to search the graph level by level.

Stage	Description
Step 1: Create Graph	Represent campus locations as vertices and paths as edges.
Step 2: Initialize BFS	Start from the selected location and add it to the queue.
Step 3: Explore Neighbors	Visit connected locations level by level.
Step 4: Track Parent	Store the previous location for every visited location.
Step 5: Find Destination	Stop when the destination is reached.
Step 6: Reconstruct Path	Follow the parent information backwards to obtain the route.
Output	Display the shortest route and hop count.

7. WHY BFS FINDS THE SHORTEST PATH

BFS visits nodes level by level. The starting location is at Level 0. Locations that are one hop away are at Level 1, locations that are two hops away are at Level 2, and so on. Therefore, when BFS first reaches the destination, it has used the minimum possible number of edges.

Example:

Gate → Library → Canteen → Lab

Shortest Route: Gate → Library → Canteen → Lab

Hop Count: 3

8. RESULTS

The system successfully:

- Represents the campus as an unweighted graph.
- Finds the shortest route using BFS.
- Displays the complete path.
- Calculates the number of hops.
- Demonstrates that BFS produces the fewest-edge path.

9. CONCLUSION

The Campus Route Finder Using BFS demonstrates the practical use of graph traversal in finding shortest routes. BFS is suitable for an unweighted campus graph because it explores locations level by level and guarantees a path with the minimum number of edges. The system provides a simple and effective way to find routes between campus locations.

10. REFERENCES

1. Project presentation: "Campus Route Finder Using Breadth-First Search (BFS)", Innovative Assignment – I, Foundations of Artificial Intelligence.

2. Project GitHub repository: https://github.com/PRAVEEN14031/FAI-1/tree/main/project-name